***New Phytologist* Supporting Information**

Article title: Parent-of-origin specific allelic expression in outbreeding *Arabidopsis arenosa* identifies antagonistic parental enrichment in protein degradation pathways

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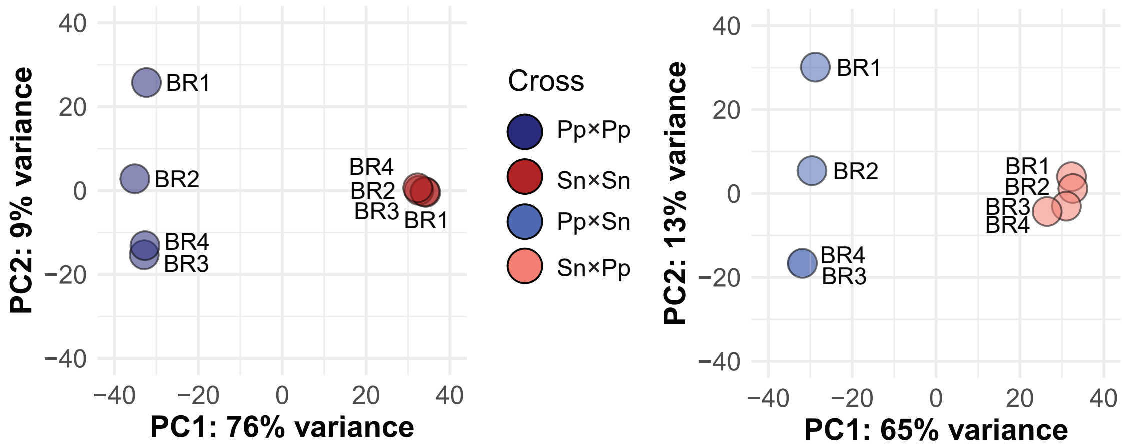
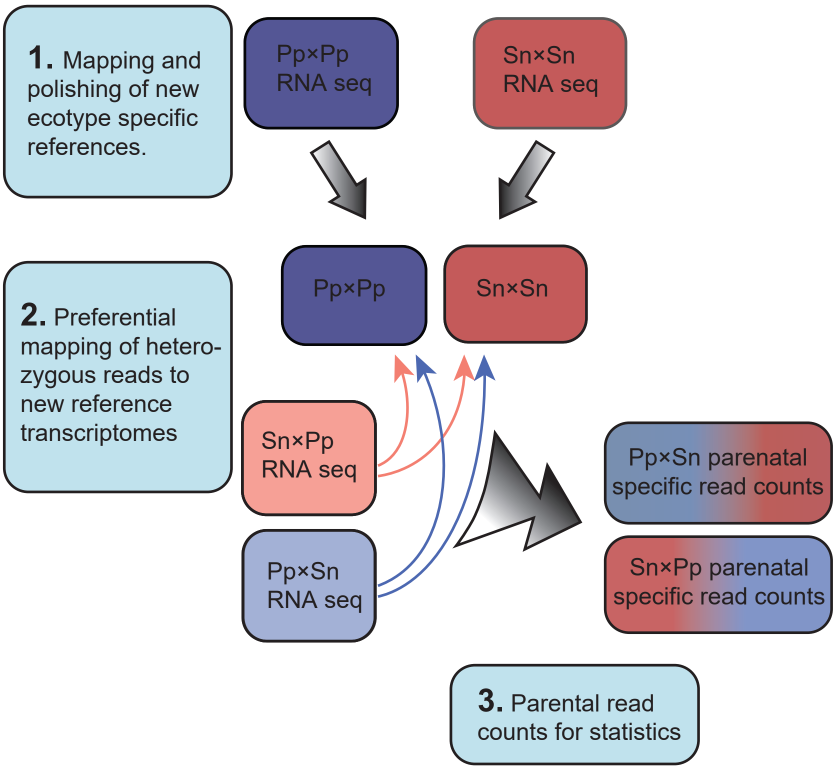
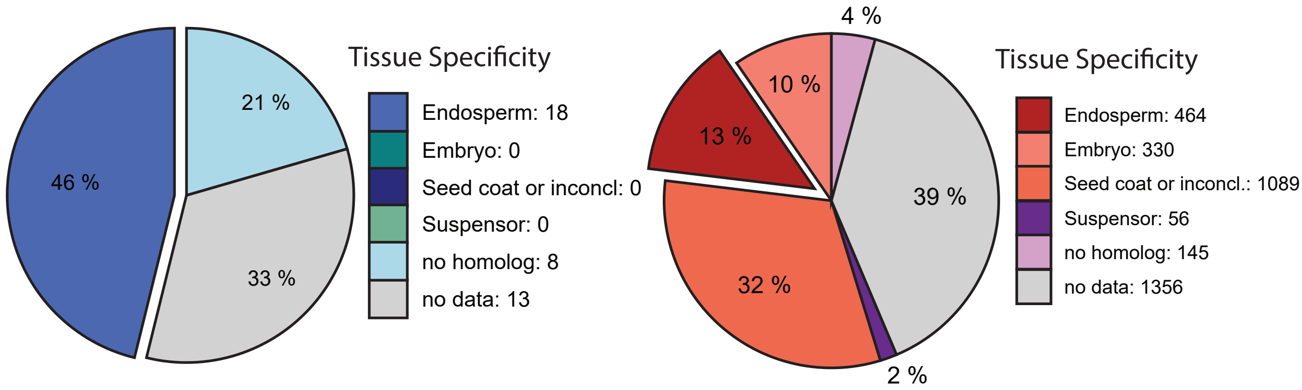


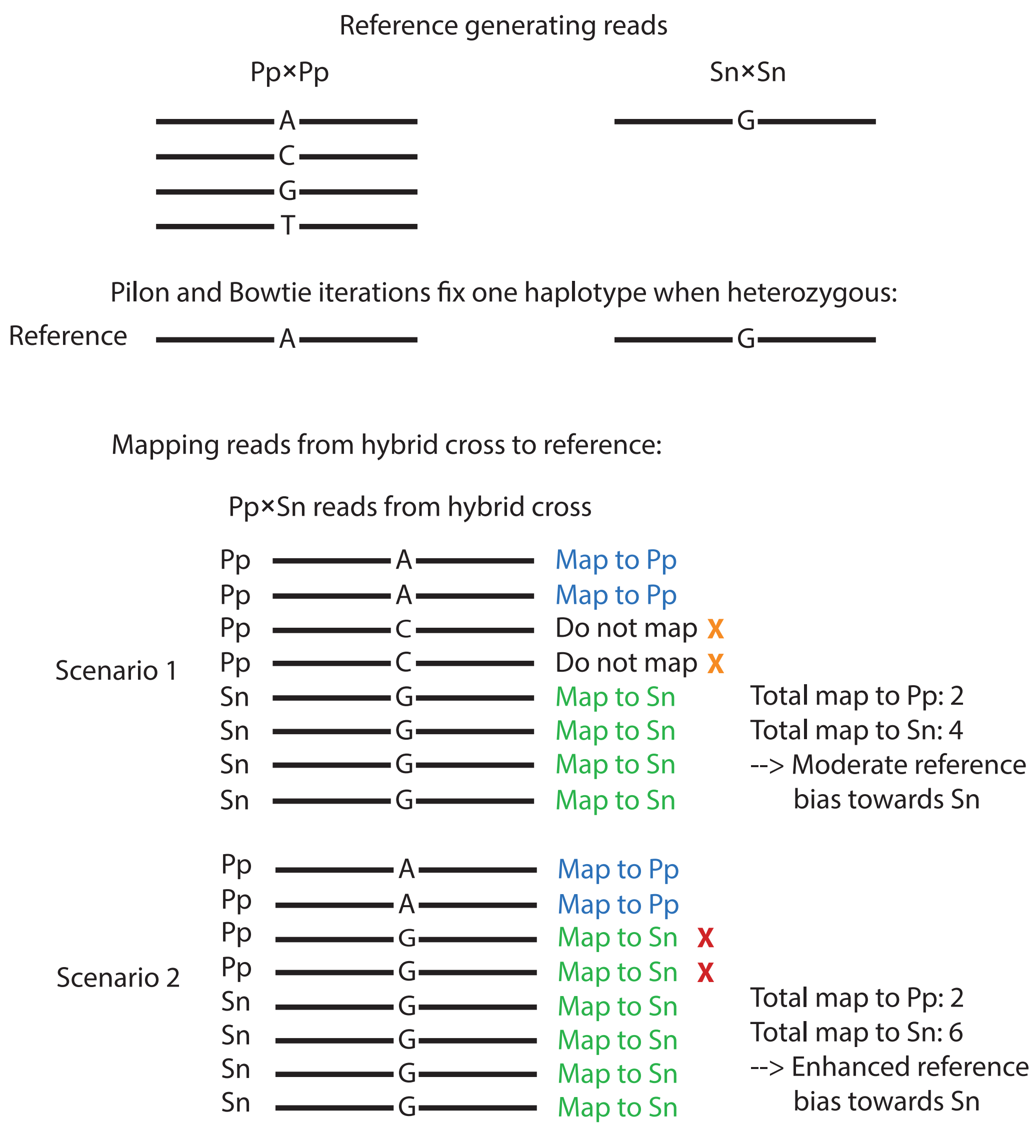
Supplementary Figure S1. Principal component (PC) analysis of mapped reads from four biological replicates (BR) from each intra-accession crosses (left) and inter-accession crosses (right) within and between Pusté Pole (Pp) and Strecno (Sn) accessions of *Arabidopsis arenosa.*



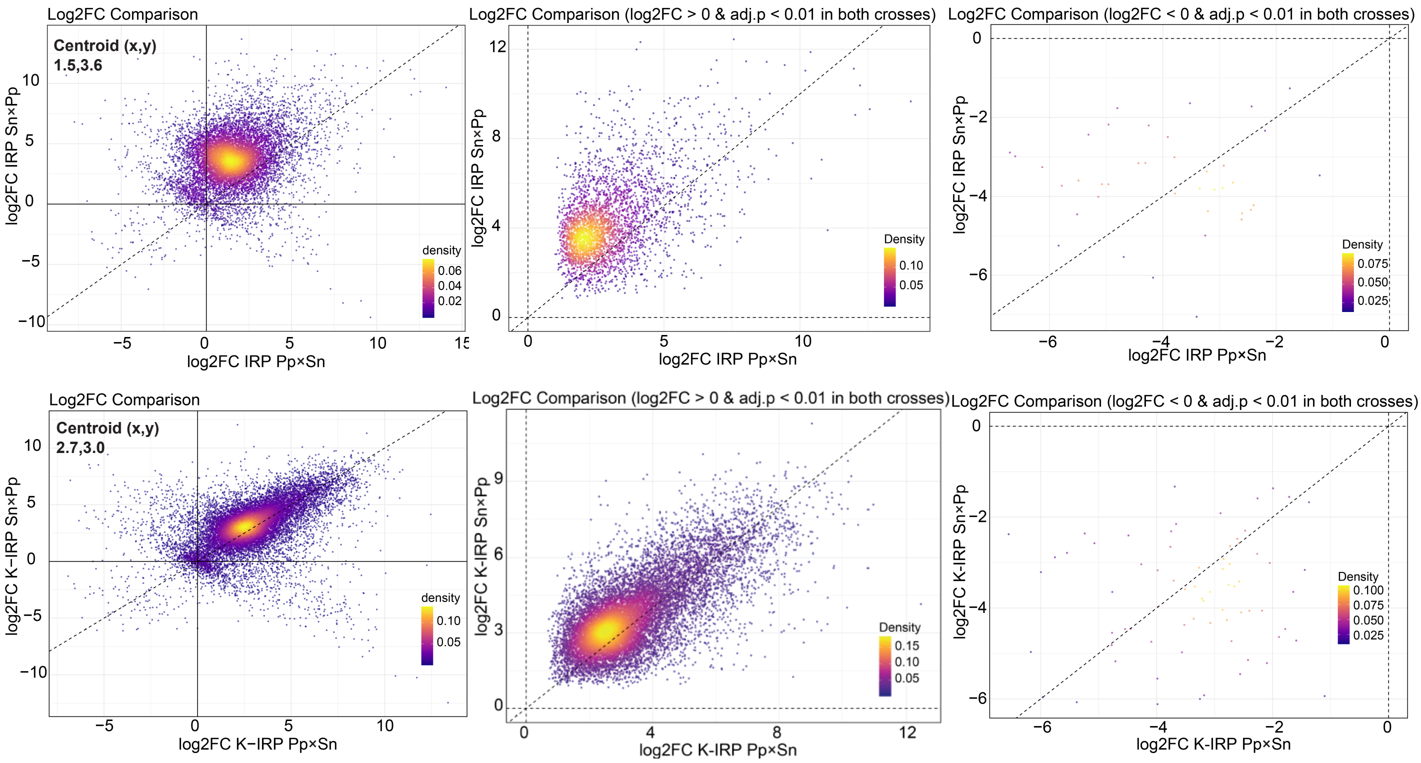
Supplementary Figure S2. Outline of the informative reads pipeline (IRP). RNA sequencing reads from *Arabidopsis arenosa* intra-accession crosses of accessions Puste Pole (Pp) and Strecno (Sn) (Pp×Pp and Sn×Sn) were used in the Informative read pipeline (IRP) to generate and polish accession specific transcriptomes. RNA sequencing reads from inter-accession crosses (Pp×Sn and Sn×Pp) were subsequently mapped to the new transcriptomes following the IRP pipeline to identify parent of origin dependent expression patterns.



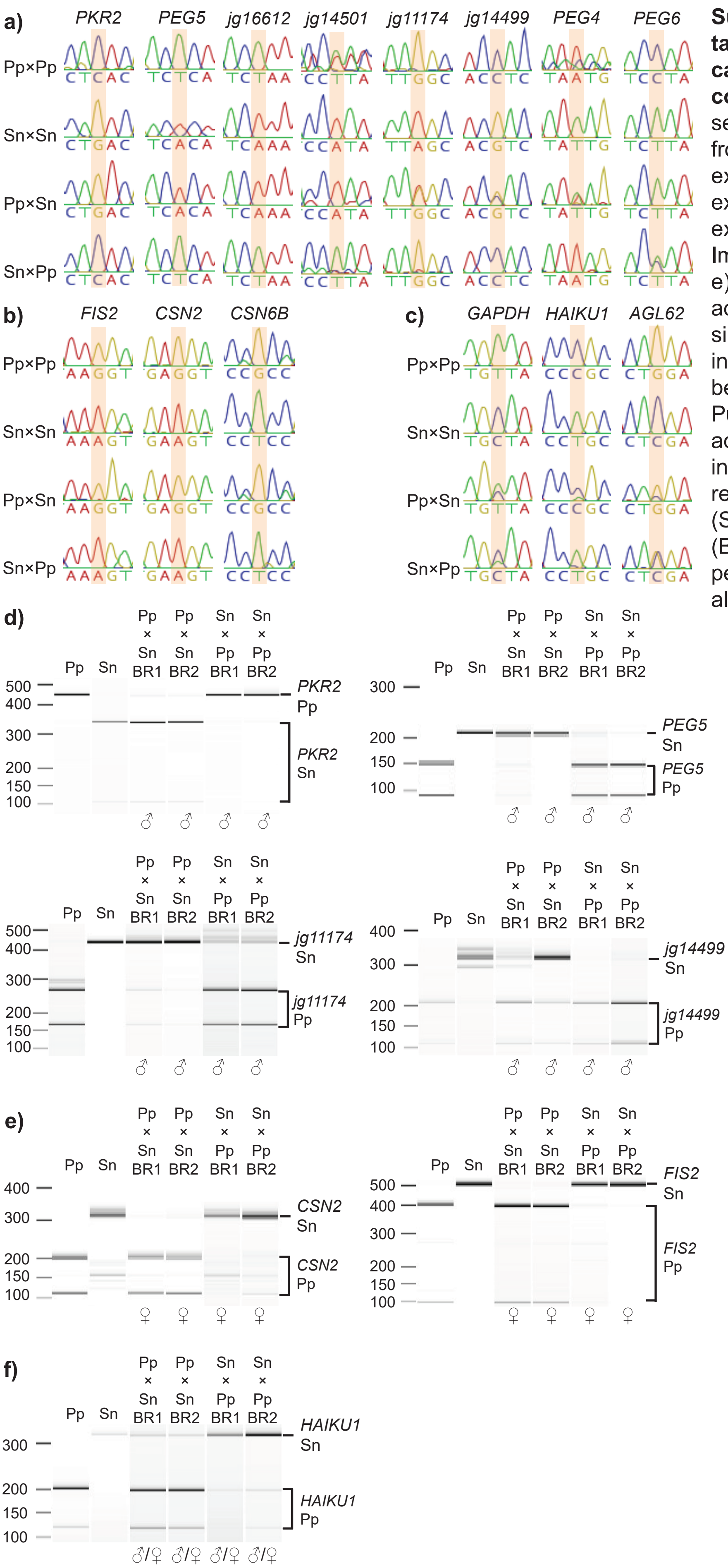
Supplementary Figure S3. Tissue enrichment IRP set. Paternally biased genes are confirmed to be endosperm specific analyzing *A. thaliana* homologs of the paternally biased genes identified in *A. arenosa* in published data (blue). Maternal biased expression is in a large part shown to be seed coat expressed genes analyzing published data for *A. thaliana* homologs of the maternally biased genes identified in *A. arenosa*.



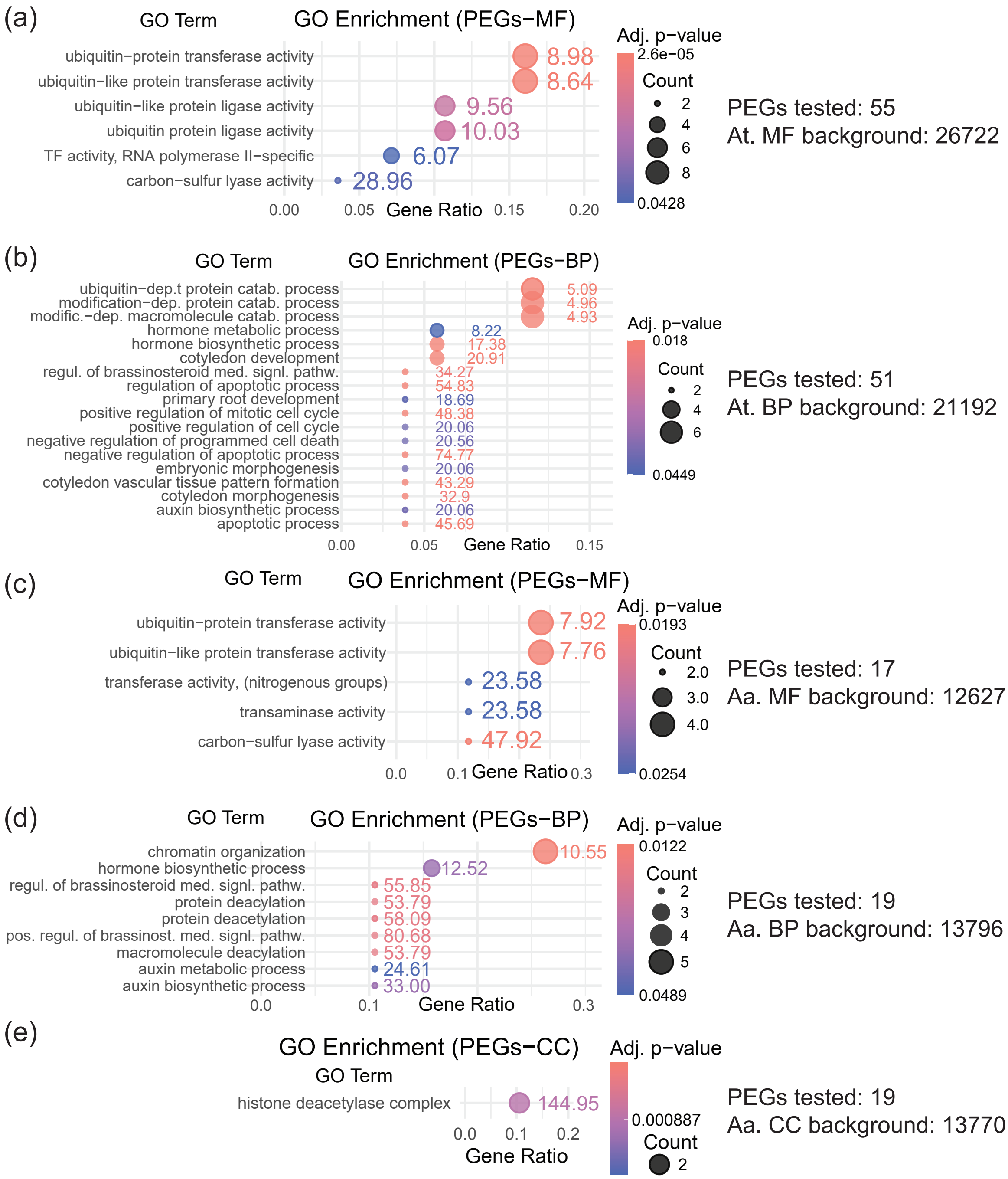
Supplementary Figure S4. Mapping behavior of sequencing reads derived from heterozygous populations mapped to a fixed reference transcriptome. Intrapopulation crosses were sequenced to generate reads to make two new population specific references, designated as Puste Pole reference Pp×Pp and Strecno reference Sn×Sn. At a particular single nucleotide polymorphism (SNP) position, reference Pp×Pp is postulated to be heterozygous with two to four possible haplotypes (A, C, G, or T). However, due to the fixation of only one haplotype during Pilon-Bowtie iterations, the reference transcriptome for Pp×Pp used in the final analysis represents in this example, the allele 'A' at this SNP position. Conversely, Reference Sn×Sn is homozygous at the same SNP position, consistently showing the allele 'G'. The sequencing reads utilized in this experiment are derived from a hybrid of the two references, containing a mix of alleles from both population Pp and population Sn. In this example two separate scenarios are presented that will lead to a map bias towards the Sn×Sn reference to a varying degree, while the expected map ratio should be 50/50 in the hybrid.



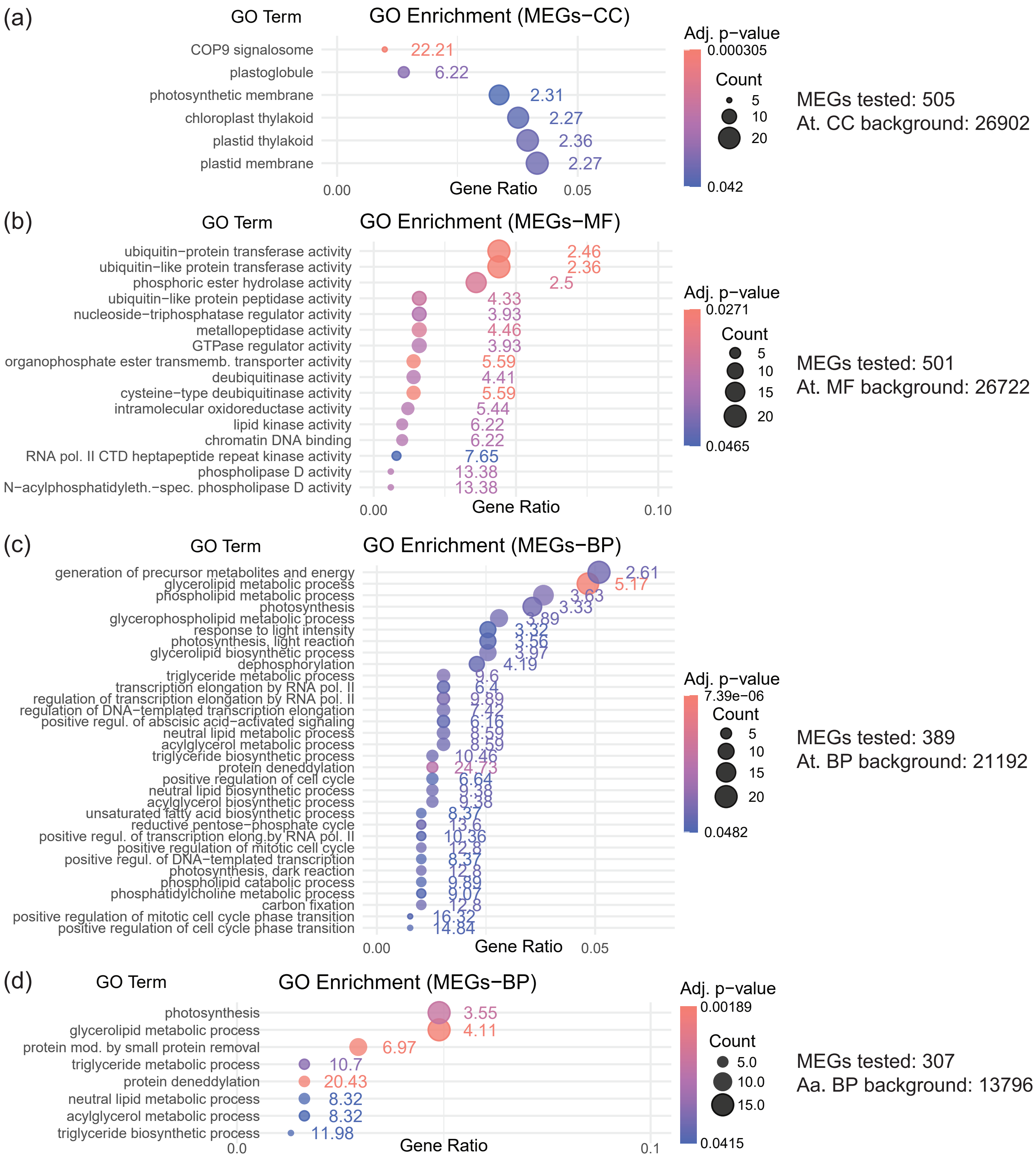
Supplementary Figure S5. Density plot of Log2 Fold Change (Log2FC) values from reciprocal inter-accession crosses between Strecno (Sn) and Pusté Pole (Pp). Log2FC values were plotted against each other, and estimated point density is indicated by colors (blue indicate low density, yellow indicate higher density). Top three panels show density of the informative read pipeline (IRP) Log2FCs, while bottom panels show density of the k-mer-informative read pipeline (K-IRP) LogFCs. From left to right the panels show: full data, significantly maternally biased genes, and significantly paternally biased genes. Neither of the full distributions are Gaussian (Kolmogorov-Smirnov test, $p<0.01$), making the t-test inappropriate. The difference between the two full data distributions is significant (Mann-Whitney U-test, $p<0.01$). The dotted diagonal line is added as a reference line of equality (y=x).



Supplementary Figure S6. Experimental validation with RT-PCR for selected candidate imprinted genes and controls. a-c) Chromatograms of Sanger sequenced PCR products from cDNA from indicated crosses. a) Paternally expressed genes (PEGs), b) Maternally expressed genes (MEGs), c) Biparentally expressed genes (BEGs) (controls). d-f) Imprinting analysis of selected d) PEGs, e) MEGs and f) control BEGs using accession specific restriction digest in single nucleotide polymorphisms (SNPs) in controls and reciprocal crosses between accessions Strecno (Sn) and Pusté Pole (Pp). For each panel the accession specific digestion pattern is indicated. Bioanalyzer images of one replicate for controls Pp (Pp × Pp) and Sn (Sn × Sn) and two biological replicates (BR) for heterozygous crosses are shown per target gene. Maternal parent is always indicated first/top in a cross.

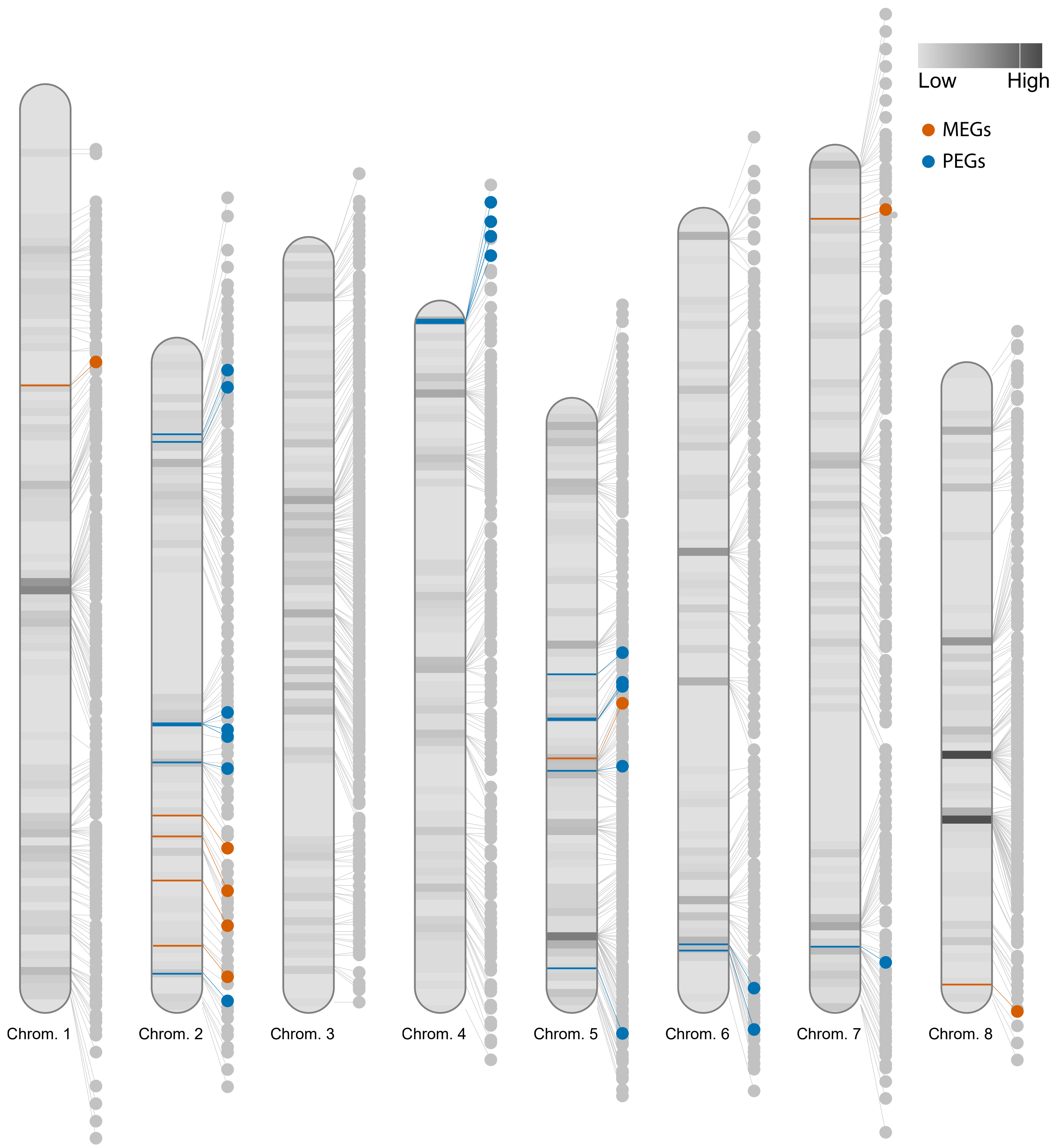


Supplementary Figure S7. Gene Ontology (GO) enrichment analysis of *A. thaliana* homologs of *A. arenosa* paternally expressed genes (PEGs). a-b) *A. thaliana* (At) GO categories and genome was used as background for the primary GO enrichment analysis. Annotation data sets used were Molecular function (MF), Biological Process (BP). No significant enrichment was found for PEGs using the Cellular component (CC) annotation set. c-e) EggNOG was used to GO annotate the *A. arenosa* (Aa) genome in a secondary GO enrichment analysis. Numbers next to the points inside plots represent the Odds enrichment ratio. Size of the point reflects the number of genes in the test list and Gene Ratio the fraction of total test list. Adjusted p-values reflect significance from Fisher's exact test with Benjamini Hochberg correction for multiple testing. Numbers of genes tested and total numbers of genes in each GO reference background is noted to the right.



Supplementary Figure S8. Gene Ontology (GO) enrichment analysis of *A. thaliana* homologs of *A. arenosa* endosperm enriched maternally expressed genes (MEGs). a-c) *A. thaliana* (At) GO categories and genome was used as background for the primary GO enrichment analysis. Annotation data sets used were Molecular function (MF), Biological Process (BP) and Cellular component (CC). d) EggNOG was used to GO annotate the *A. arenosa* (Aa) genome in a secondary GO enrichment analysis. No significant enrichment was found for MEGs using the MF annotation set, the full CC-enrichment set is presented in main Figure 5 of the manuscript. Numbers next to the points inside plots represent the Odds enrichment ratio. Size of the point reflects the number of genes in the test list and Gene Ratio the fraction of total test list. Adjusted p-values reflect significance from Fisher's exact test with Benjamini Hochberg correction for multiple testing. Numbers of genes tested and total numbers of genes in each GO reference background is noted to the right.

Supplementary Figure S9. Phylogenetic representation of the F-box family in *A. arenosa* and *A. thaliana*. Maximum-likelihood tree of F-box domain sequences. Coloured boxes indicate predominant C-terminal motif. Tip colours denote imprinting status, with paternally expressed genes in blue and maternally expressed genes in orange. Diamonds on branches represent support values (UFboot/SH-aLRT): black diamonds indicate high support (>95/80), and grey indicates moderate support (90–95 / 70–80). When only one support value exceeds the threshold, the other is shown as "–". Branches without a displayed diamond or value received no support above the reporting thresholds. Scale bar indicates the number of substitutions per site.



Supplementary Figure S10. Ideogram illustrating the genomic distribution of F-box genes across the eight chromosomes of *A. arenosa*. Grey shading indicates gene density in 20 kb windows, calculated from gene midpoints. Non-imprinted genes are marked in grey, paternally expressed genes (PEGs) in blue, and maternally expressed genes (MEGs) in orange at their genomic positions.